

W4S4T1

Due: 18/6/2026

Sample Collection: students

Use the following sample data for all tasks.

```
db.students.insertMany([
```

```
{
```

```
  student_id:1,
```

```
  name:"Saran",
```

```
  department:"CSE",
```

```
  city:"Bangalore",
```

```
  cgpa:8.5,
```

```
  age:21,
```

```
  gender:"Male"
```

```
},
```

```
{
```

```
  student_id:2,
```

```
  name:"Priya",
```

```
  department:"ECE",
```

```
  city:"Hyderabad",
```

```
  cgpa:9.0,
```

```
  age:22,
```

```
  gender:"Female"
```

```
},
```

```
{
```

```
  student_id:3,
```

```
  name:"Rahul",
```

```
  department:"CSE",
```

```
  city:"Chennai",
```

```
cgpa:7.8,  
age:21,  
gender:"Male"  
},  
{  
student_id:4,  
name:"Deepa",  
department:"EEE",  
city:"Bangalore",  
cgpa:8.9,  
age:22,  
gender:"Female"  
},  
{  
student_id:5,  
name:"Anjali",  
department:"ECE",  
city:"Pune",  
cgpa:8.2,  
age:20,  
gender:"Female"  
},  
{  
student_id:6,  
name:"Kumar",  
department:"Mechanical",  
city:"Hyderabad",  
cgpa:7.5,  
age:23,
```

gender:"Male"

},

{

student_id:7,

name:"Sneha",

department:"CSE",

city:"Bangalore",

cgpa:9.1,

age:21,

gender:"Female"

},

{

student_id:8,

name:"Varun",

department:"Civil",

city:"Chennai",

cgpa:8.0,

age:22,

gender:"Male"

},

{

student_id:9,

name:"Charan",

department:"Mechanical",

city:"Pune",

cgpa:7.2,

age:23,

gender:"Male"

},

```
{  
  student_id:10,  
  name:"Harsha",  
  department:"CSE",  
  city:"Hyderabad",  
  cgpa:8.7,  
  age:21,  
  gender:"Male"  
}  
]);
```

COUNT Aggregation Tasks

Task 1

Find the total number of students.

Task 2

Find the number of students in each department.

Task 3

Find the number of students in each city.

Task 4

Find the number of male students.

Task 5

Find the number of female students.

AVG Aggregation Tasks

Task 6

Find the average CGPA of all students.

Task 7

Find the average CGPA department-wise.

Task 8

Find the average age department-wise.

Task 9

Find the average CGPA city-wise.

Task 10

Find the average age of all students.

SUM Aggregation Tasks**Task 11**

Find the sum of all CGPAs.

Task 12

Find the total age of all students.

Task 13

Find the total CGPA department-wise.

Task 14

Find the total age city-wise.

Task 15

Find the total CGPA of CSE students.

MAX Aggregation Tasks**Task 16**

Find the highest CGPA.

Task 17

Find the highest age.

Task 18

Find the highest CGPA department-wise.

Task 19

Find the highest age city-wise.

Task 20

Find the department having the maximum average CGPA.

MIN Aggregation Tasks**Task 21**

Find the minimum CGPA.

Task 22

Find the minimum age.

Task 23

Find the minimum CGPA department-wise.

Task 24

Find the minimum age city-wise.

Task 25

Find the city having the lowest average CGPA.

GROUP BY Equivalent (\$group)**Task 26**

Group students by department and count them.

Task 27

Group students by city and count them.

Task 28

Group students by gender and count them.

Task 29

Group students by department and find average CGPA.

Task 30

Group students by city and find average age.

MATCH + GROUP**Task 31**

Find the number of students with CGPA greater than 8.

Task 32

Find the average CGPA of CSE students.

Task 33

Find the total number of female students.

Task 34

Find the highest CGPA among male students.

Task 35

Find the lowest CGPA among female students.

SORT Aggregation

Task 36

Display departments ordered by average CGPA in descending order.

Task 37

Display cities ordered by student count.

Task 38

Display departments ordered by maximum CGPA.

Task 39

Display cities ordered by average age.

Task 40

Display departments ordered by number of students.